UCSD ROCKET PROPULSION LABORATORY • 2024–2025

Daedalus | Structures Lead

Engineering objective

Design and manufacture a student rocket; 4,086 ft was the historical simulated apogee.

My structures scope

Nose geometry, shoulder/body fit, recovery attachment and structural integration.

Manufacturing and integration

Supported PLA structural manufacture, mechanical fit checks and hardware integration with the team.

Native assembly • reference envelope



25 in
tip to aft body

1.5 in
body diameter

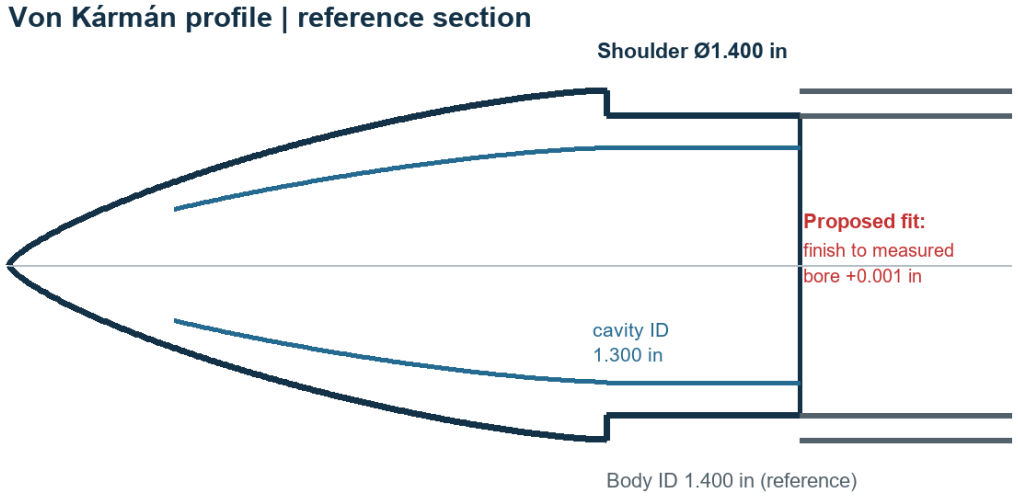
6 in nose

Status Manufactured and ground-tested; never flown. Performance figures are simulations.

Assembly image and dimensions are an October 2026 reference reconstruction, not as-built measurements.

What carries the deployment load?

NOSE / BODY FIT

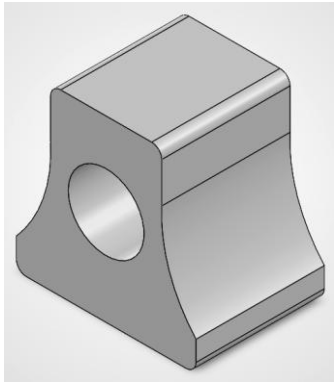


$$t_{r,\,\mathrm{ref}} = \frac{1.400 - 1.300}{2} = 0.050 \text{ in}$$

$$C_{\mathrm{fit}} = 1.400 - 1.401 = -0.001 \text{ in}$$

Finish to measured bore +0.001 in; qualify with 15–20 N pull and ≥35 N ejection.

ROUNDED EYELET / LOAD PATH

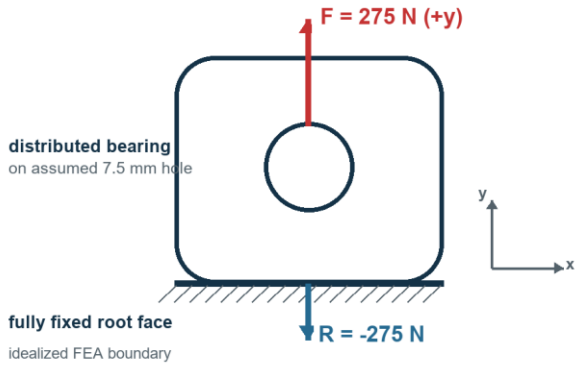


0.7 × 0.7 × 0.5 in

Reference envelope

7.5 mm hole

Retrospective assumption



CONDITIONAL RECOVERY SCREEN

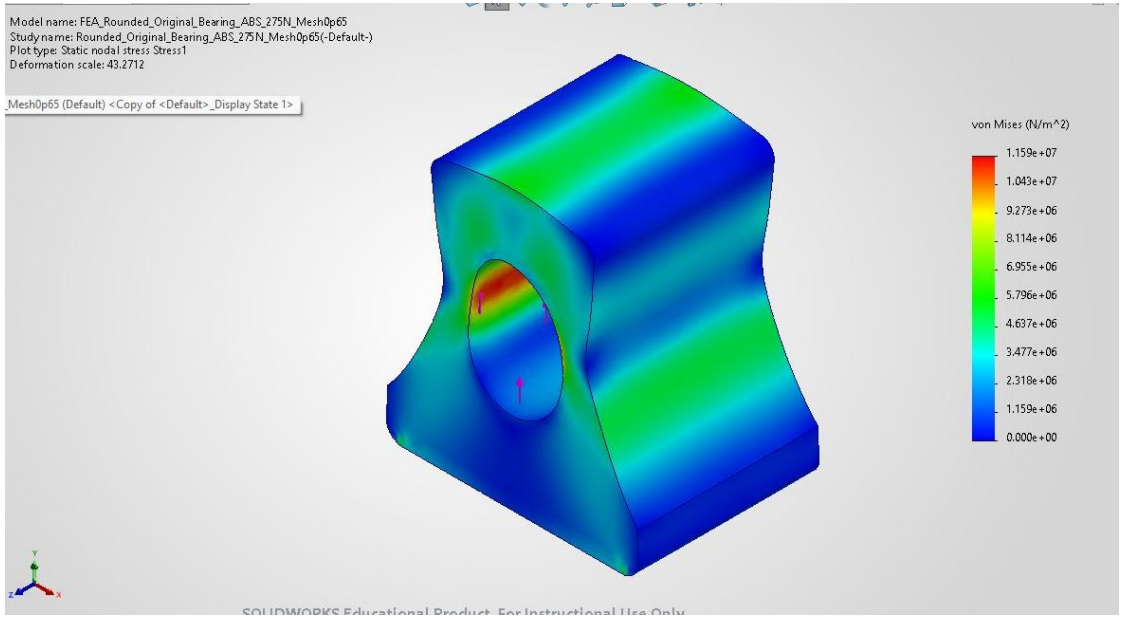
$$F_{\mathrm{open}} = \tfrac{1}{2}\rho V^2 C_D A K_o = 135.5 \text{ N}$$

$$F_{\mathrm{screen}} = K_s F_{\mathrm{open}} = 271 \text{ N} \Rightarrow 275 \text{ N}$$

D = 0.6096 m; ρ = 1.225 kg/m^3; Cd = 1.75; t = 1.5 s; opening and screen multipliers = 2 each.

Wider rounded eyelet lowers modeled stress

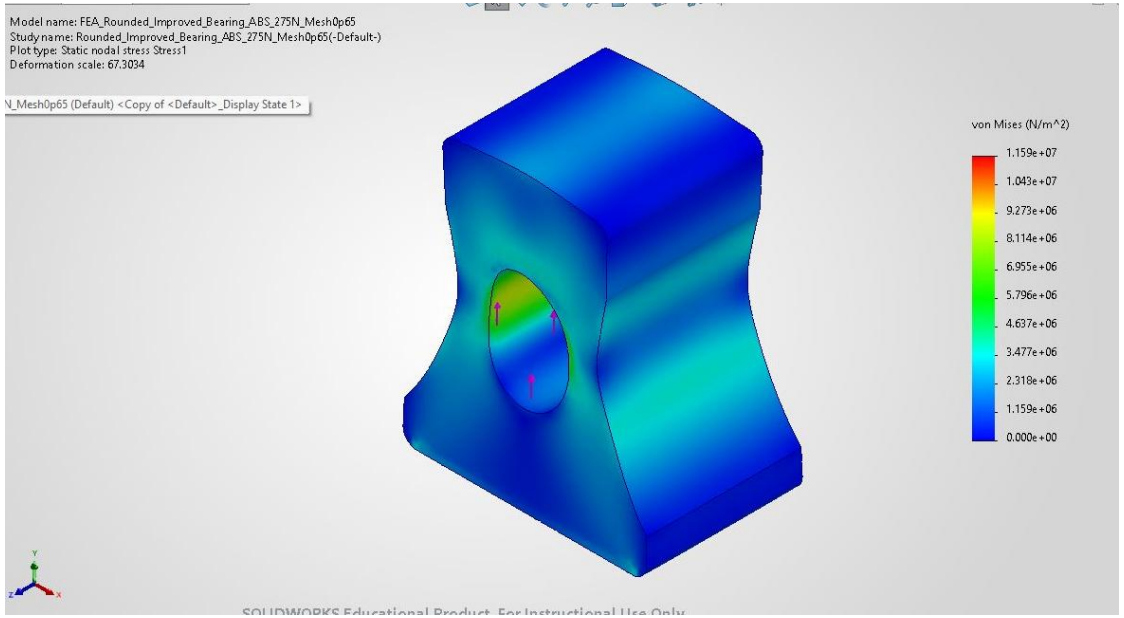
REFERENCE RECONSTRUCTION



11.591 MPa → 8.869 MPa

−23.5% peak stress

WIDER CONCEPT



0.04138 → 0.03299 mm

−20.3% max displacement

Wider net section and top ligament spread the hole load; added material and an unmodeled root are the tradeoffs.

Common scale 0–11.591 MPa. Conditional 275 N; fixed root; ABS. Printed PLA and dynamics unvalidated.